Computer Lab 1: High Dimensional Data Analysis

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Part 0: Matrix Algebra and Random Vectors

We have the matrix $\mathbf{L}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\mathbf{L} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

i)

We are tasked with computing the transpose, determinant, rank, and trace of L. L is symmetric so the transpose of L, L^T is simply L. The determinant is $|L| = 11$, the rank is 2, and the trace is 7.

ii)

The determinant is not equal to 0 and $\mathbf{L}$ is thus invertible. The inverse $\mathbf{L}^{-1}$ is

$$\mathbf{L}^{-1} = \begin{pmatrix} 0.27272727 & -0.09090909 \\ -0.09090909 & 0.36363636 \end{pmatrix}$$

iii)

We are tasked with performing the matrix multiplication of $\mathbf{L}$ and $\mathbf{u}$:

$$\mathbf{Lu} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

iv)

The eigenvalues and eigenvectors of $\mathbf{L}$ are:

Eigenvalues:

$$\lambda_1 = 4.618034, \qquad \lambda_2 = 2.381966$$

Eigenvectors (columns correspond to the respective eigenvalues):

$$V = \begin{pmatrix} -0.8506508 & 0.5257311 \\ -0.5257311 & -0.8506508 \end{pmatrix}$$

v)

The quadratic form of $\mathbf{u}$ with respect to $\mathbf{L}$ is:

$$\mathbf{u}^\top \mathbf{L} \mathbf{u} = 15$$

vi)

The inner product of $\mathbf{u}$ and $\mathbf{v}$, $(\mathbf{u}^\top \mathbf{v})$ is 5.

Part 1: Iris Dataset

In this part we will use the `iris` dataset to see aspects of multivariate analysis.

i)

Table 1, below, displays descriptive statistics for the numerical variables in the dataset, while Table 2 shows the frequency of the categorical variable, *species*.

Table 1: Descriptive statistics of the Iris dataset

	Variable	Mean	SD	Min	Max	Median
Sepal.Length	Sepal.Length	5.843333	0.8280661	4.3	7.9	5.80
Sepal.Width	Sepal.Width	3.057333	0.4358663	2.0	4.4	3.00
Petal.Length	Petal.Length	3.758000	1.7652982	1.0	6.9	4.35
Petal.Width	Petal.Width	1.199333	0.7622377	0.1	2.5	1.30

Table 2: Frequency table of Species in the Iris dataset

Species	Frequency
setosa	50
versicolor	50
virginica	50

The estimated mean vector is thus:

$$\bar{\mathbf{x}} = \begin{pmatrix} 5.843333 \\ 3.057333 \\ 3.758000 \\ 1.199333 \end{pmatrix}$$

ii)

Figure 1 below, displays a scatter plot matrix for the numerical variables. We see clear indications of a strong positive correlations between several of the variable pairs.

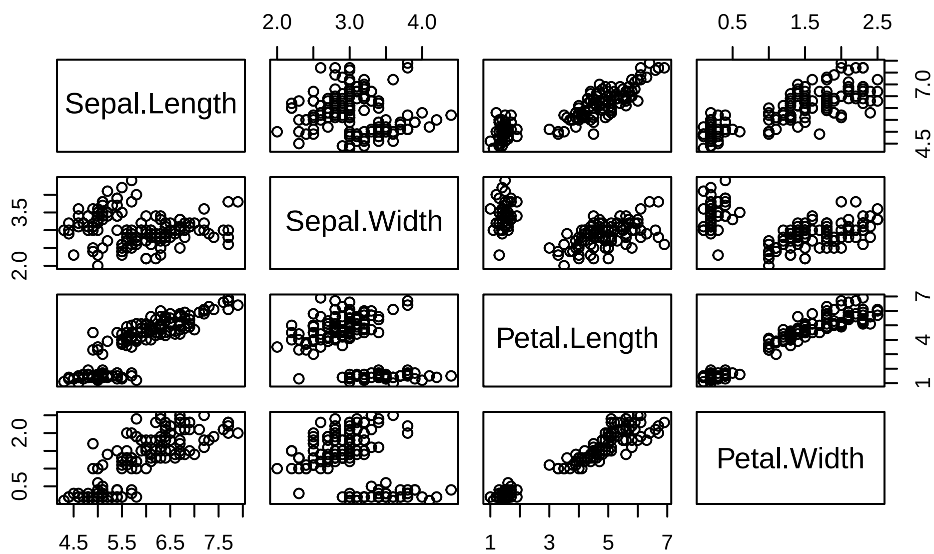

Figure 1: Scatterplot matrix of the numeric variables in the Iris dataset

The correlation matrix, illustrated in Table 3, below, indeed confirms that this is the case, with strong positive correlations between *Petal Length*, *Petal With* and *Sepal Length*

Table 3: Correlation matrix of the numeric variables in the Iris dataset

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Sepal.Length	1.0000000	-0.1175698	0.8717538	0.8179411
Sepal.Width	-0.1175698	1.0000000	-0.4284401	-0.3661259
Petal.Length	0.8717538	-0.4284401	1.0000000	0.9628654
Petal.Width	0.8179411	-0.3661259	0.9628654	1.0000000

iv)

Figure 2, below, plots histograms of the numeric variables in the dataset. We see that the distribution for *Sepal Width* looks the most symmetric, centered around its mean of approximately 3. *Sepal Length* also appears moderately symmetric but has a much less “normal-looking” distribution, as it is noticeably flatter than a typical bell-shaped curve. The *Petal Length* and *Petal Width* distributions, in contrast, appear almost bimodal, with there being one group with similarly small petals, as well as one group with a variety of larger petal sizes.

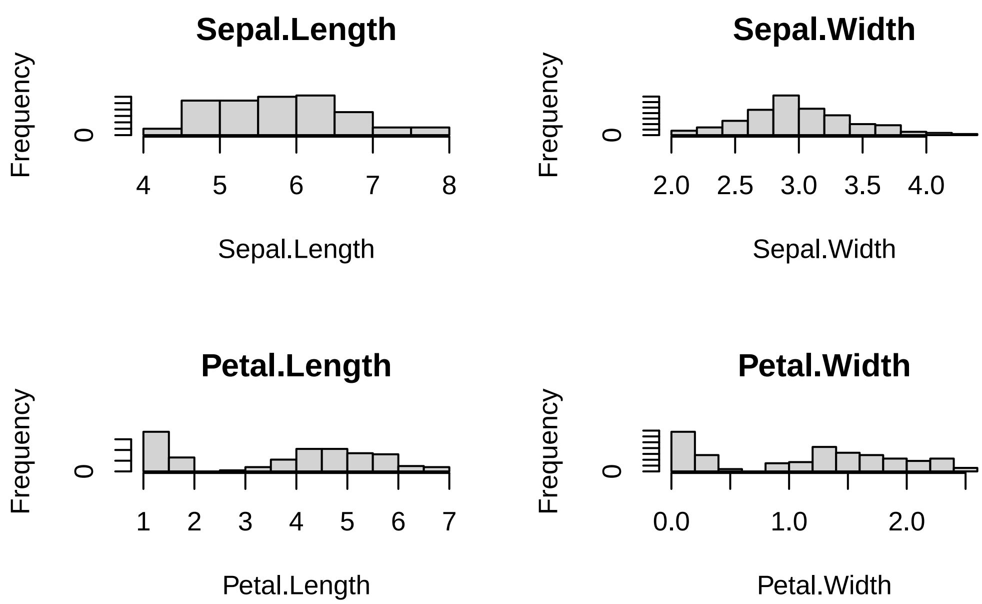

Figure 2: Histograms of the numeric variables in the Iris dataset

Part 2: Weight and Length of Newborn Children

In this section we will analyze data consisting of the height (in cm) and weight (in grams) of 736 newborn children.

i)

The estimated mean-vector $\bar{\mathbf{x}}$, containing the average *weight* and *length* of the infants in the dataset is:

$$\bar{\mathbf{x}} = \begin{pmatrix} 3233.545 \\ 49.238 \end{pmatrix}$$

The estimated covariance matrix $\mathbf{S}$ is illustrated in Table 4, below. As expected there is positive covariance between *length* and *weight*.

Table 4: Estimated covariance matrix of Weight and Length

	Weight	Length
Weight	220276.658	915.296
Length	915.296	4.443

Figure 3, on the following page, illustrates Histograms and Q-Q plots of the two variables in the datasets. Both plot types indicate that the marginal distributions adhere well to the normal distribution.

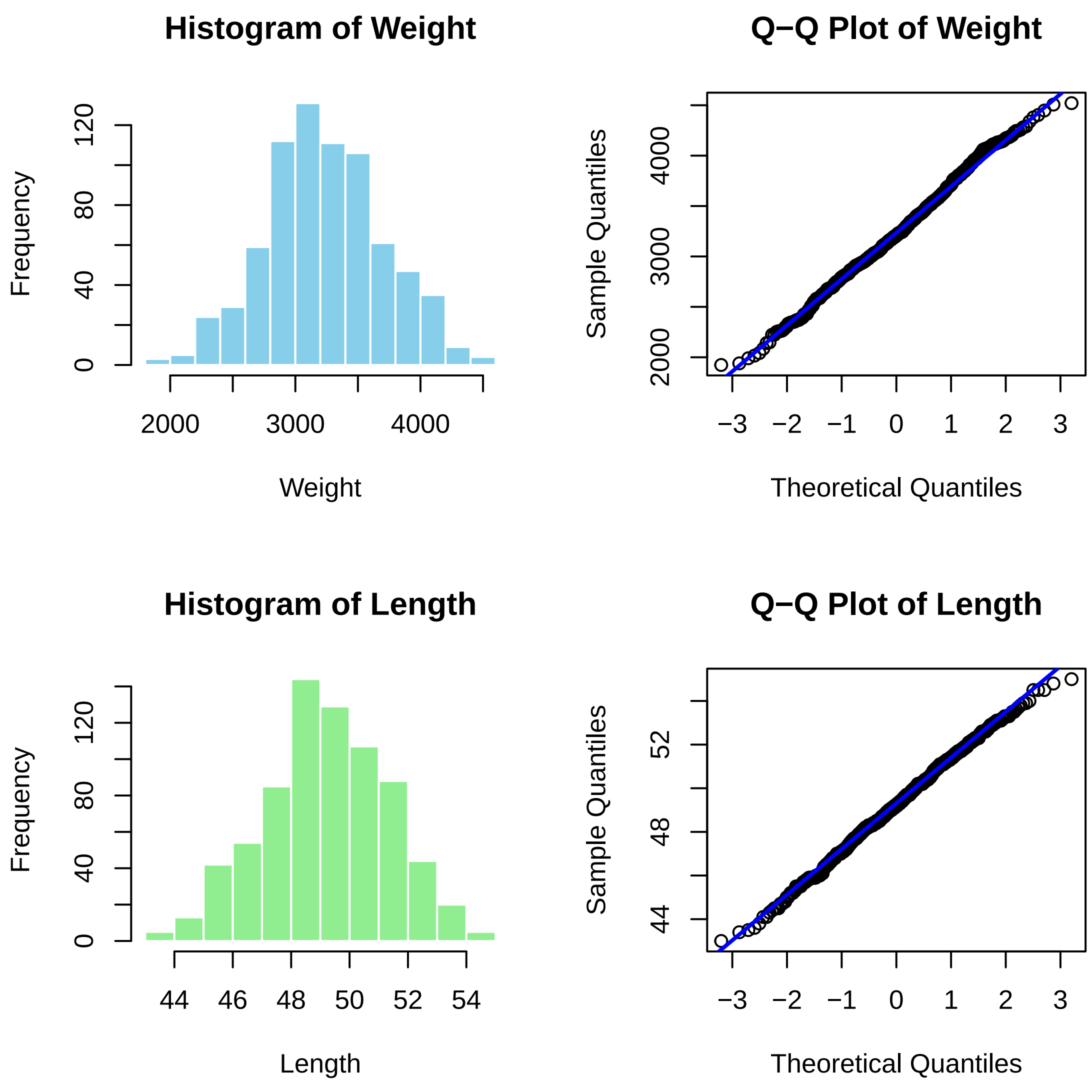

Figure 3: Histograms and Q-Q plots of Weight and Length

Figure 4, below, once again displays a strong positive correlation between the *weight* and *height*.

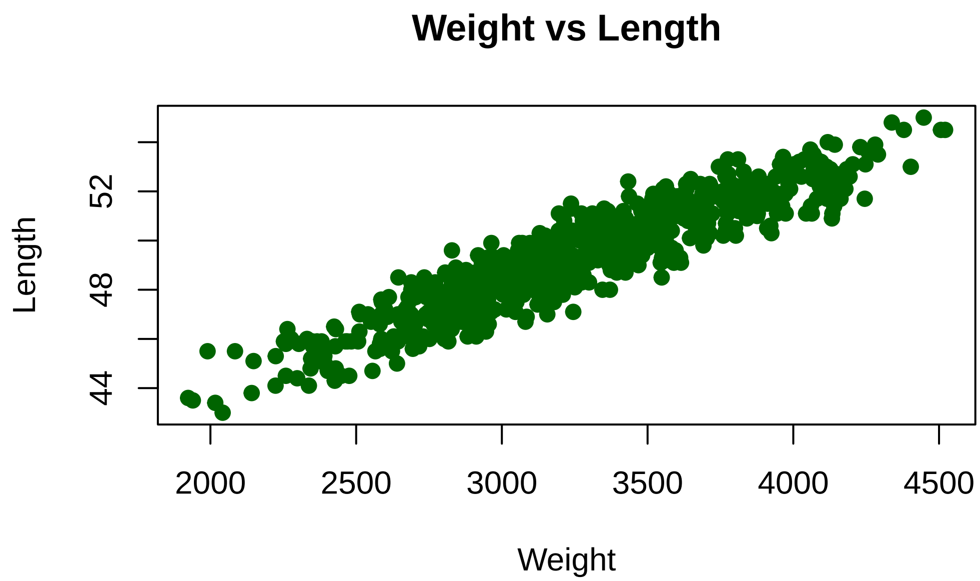

Figure 4: Scatter plot of Weight vs Length

iii-iv)

Now we want to consider the following score system:

- **Score 0:** If the weight and length fall outside 95% of the typical values for the population.
- **Score 1:** If the measurements fall in the category between 75% and 95% of the typical values.
- **Score 2:** In all other cases (i.e., inside the 75% region)

We want to draw the ellipsoids that serve as the classification regions for the scores, as well as classify the children in the dataset. This is illustrated in Figure 5, on the following page. In red we have the ellipsoid corresponding to the estimated multivariate distribution of the data, centered at the mean vector. The axes point along the eigenvectors of the covariance matrix, and their lengths are proportional to the square roots of the eigenvalues. The points are then classified by first computing their squared Mahalanobis distance for each point $\mathbf{x}_i$

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

Points with d_i^2 larger than the 95% threshold derived from the F-distribution with p and $n-p$ degrees of freedom, 6.02, are classified as Score 0. Points with d_i^2 between the 75% and 95% thresholds, 2.78 – 6.02, are assigned Score 1, and points with d_i^2 smaller than the 75% threshold, 2.78, are assigned Score 2.

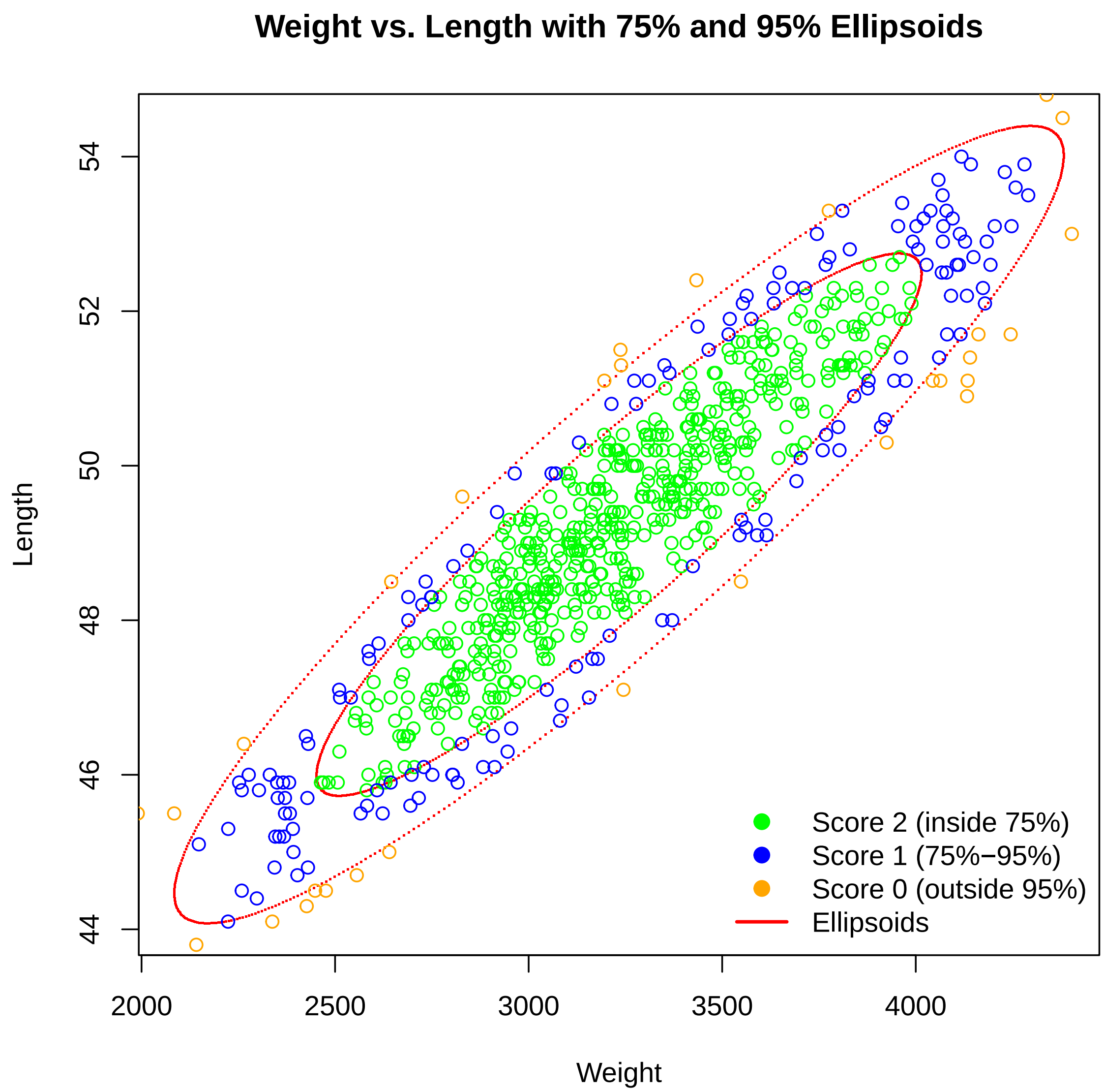

Figure 5: Scatter plot of weight vs. length measurements with 75% (inner) and 95% (outer) confidence ellipses. Points are colored according to a three-level scoring system

In total, Score 0 includes 37 children (5%), Score 1 includes 158 children (21.5%), and Score 2 includes 541 children (73.5%), which aligns with the expected proportions given the 75% and 95% thresholds.

v)

We now consider the spectral decomposition of the covariance matrix $\mathbf{S}$, which expresses $\mathbf{S}$ in terms of its eigenvectors and eigenvalues. For a real symmetric matrix, this decomposition is

$$\mathbf{S} = \mathbf{E}\Lambda\mathbf{E}^\top$$

where $\mathbf{E}$ us an orthogonal matrix containing the eigenvectors as columns and Λ is a diagonal matrix with the corresponding eigenvalues. Thus:

$$\mathbf{E} = \begin{bmatrix} -0.99999 & 0.00416 \\ -0.00416 & -0.99999 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 220280.5 & 0 \\ 0 & 0.64005 \end{bmatrix}, \quad \mathbf{S} = \mathbf{E}\Lambda\mathbf{E}^\top$$

We now transform the data $\mathbf{X}$ according to $\mathbf{Y} = \mathbf{E}^\top\mathbf{X}$, where $\mathbf{Y}$ is an $p \times n$ matrix, with each column corresponding to an observation and each row corresponding to a new axis defined by the eigenvectors of the covariance matrix. The variability along each new axis is equal to the corresponding eigenvalue of $\mathbf{S}$, meaning that the first row of $\mathbf{Y}$ has variance equal to the largest eigenvalue, the second row has variance equal to the smaller eigenvalue. The scatter plot in Figure 6 shows the transformed data. We can see that the transformation has decorrelated the data as no more rotation is visible.

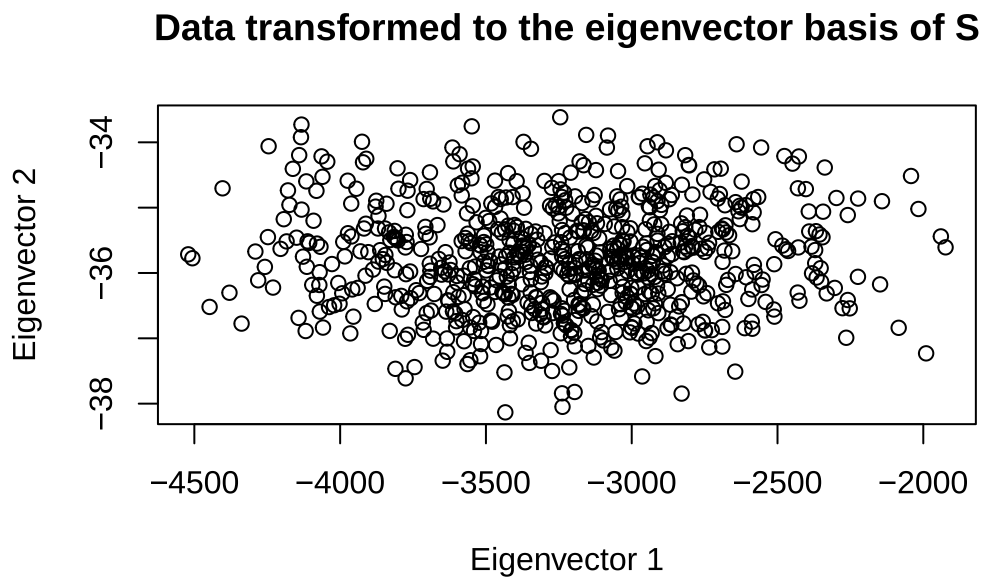

Figure 6: Scatter plot of the transformed weight and length measurements after rotation by the eigenvectors of the covariance matrix

Note, however, that in difference to regular PCA we have not standardized the data before performing the transformation. This means that the units of the variables influence the magnitude of the eigenvalues and the orientation of the axes, and that variables with larger scales will contribute more to the total variability.

Part 3: Simulation Study

We will now simulate data from a bivariate normal distribution to study the properties of the sample mean vector and sample covariance matrix. We consider the distribution $\mathbf{X}_i \sim N_2(mu, \Sigma)$ with:

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

and the sample sizes $n = \{20, 50, 200, 2000\}$. Table 5, on the following page, displays the estimated mean vector, covariance matrix, and their standard errors for each sample size. We see that the parameter vectors/matrices approach their true values as the sample size increases, which is also reflected in the decreasing standard errors.

Table 5: Estimated means, covariance matrices, and corresponding standard errors for each sample size from simulated bivariate normal data.

	n	Mean_X1	Mean_X2	SE_Mean	Cov_11	Cov_12	Cov_21	Cov_22	SE_Cov
n_20	20	0.194	0.145	0.059	1.122	0.357	0.357	1.143	0.076
n_50	50	-0.072	-0.186	0.040	0.904	0.387	0.387	0.782	0.082
n_200	200	-0.044	-0.058	0.005	0.931	0.486	0.486	0.962	0.007
n_2000	2000	0.014	0.013	0.000	1.030	0.480	0.480	0.971	0.003

Instead of simulating 1 sample of size n, we will next perform 500 simulations for each size n and estimate the MSE. The results are illustrated in Table 6 and Table 7, below.

Table 6: Average Estimated Means and SE Across 500 Repetitions

	n	X1	X2	MSE_Mean
n_20	20	-0.0004302	0.0025578	0.106
n_50	50	-0.0090867	-0.0072008	0.039
n_200	200	0.0049699	0.0013245	0.010
n_2000	2000	-0.0014314	-0.0005859	0.001

Table 7: Average SE of Covariance Matrices Across 500 Repetitions

n	MSE_Cov
20	0.345
50	0.129
200	0.034
2000	0.003

We observe that the average estimate is closer to the true value (and the bias thus smaller), for the larger sample size. In the same way the MSE also decreases with sample size. Figure 7, on the following page, plots the estimated mean vector from each of the 500 samples for all sizes n. If we consider each size n separately we see that each form an ellipsoid with the same rotation and shape, but whose axes decrease in magnitude with n.

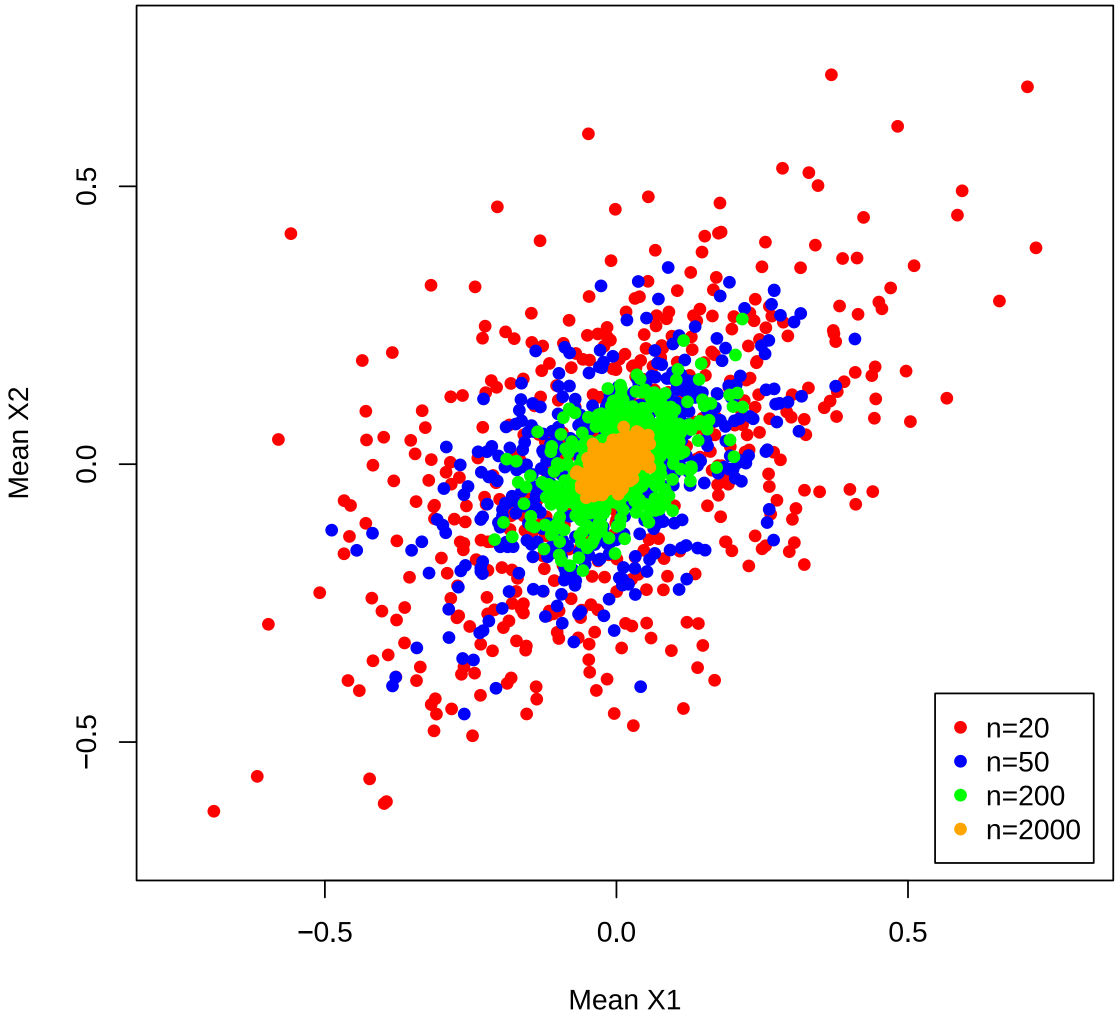

Figure 7: Simulated mean vectors for different sample sizes

We also computed the covariance matrix for $\bar{\mathbf{x}}$, for each size n across the 500 samples and compared them against the empirical covariance matrix $\frac{1}{n}\Sigma$. The difference between the estimated matrix and the empirical is illustrated in Table 7, on the following page. We observe that the difference once again approaches 0 as the sample size increases.

Table 8: Difference between Empirical and Theoretical Covariance of Means for Different Sample Sizes

SampleSize	C11	C12	C21	C22
20	0.00421	0.00143	0.00143	0.00169
50	0.00081	-0.00003	-0.00003	-0.00158
200	0.00023	0.00013	0.00013	-0.00004
2000	0.00000	0.00000	0.00000	0.00000

From the simulation result wee clearly see that the sample size affects the bias, variance and MSE of the estimators. While the expected bias is of course 0 no matter the sample size. The estimator of the mean vector, $\bar{\mathbf{x}}$ is $N_2(\mu, \frac{1}{n}\Sigma)$ which means that in practice we will observe a certain amount of bias, that will be on average smaller as the variance decreases with n. The average of 500 samples we computed for each sample size is of course normally distributed as well given that it is a linear combination 500 observations of the estimator above. It is therefore natural that although all of the estimators are unbiased to begin with (and MSE should thus be 0 for all sample sizes) we still observe nonzero MSE in our finite simulations. This reflects the sampling variability and illustrates how larger sample sizes lead to more stable and precise estimates.

Use of AI

AI (ChatGPT) has been utilized in this report primarily to assist with coding, including constructing and formatting plots, generating LaTeX code, and suggesting functions. It was also used for light spelling and grammar checks.